Chapter 03 | Functions of a real variable

600 printable exercises

Original, unofficial exercises. Source pages guide conceptual coverage only. Always check the domain assumptions.

L37 | Functions, domains and graphs

Source: PDF pages 2, 3, 4

1/5 | Apprentice

L37-P001 | Let f(x) = 2x - 4. Find f(-3).

L37-P002 | Let f(x) = 3x - 3. Find f(-1).

L37-P003 | Let f(x) = 4x - 2. Find f(1).

L37-P004 | Let f(x) = 5x - 1. Find f(3).

L37-P005 | Let f(x) = 6x + 0. Find f(-2).

L37-P006 | Let f(x) = 2x + 1. Find f(0).

L37-P007 | Let f(x) = 3x + 2. Find f(2).

L37-P008 | Let f(x) = 4x + 3. Find f(-3).

L37-P009 | Let f(x) = 5x + 4. Find f(-1).

L37-P010 | Let f(x) = 6x - 4. Find f(1).

L37-P011 | Let f(x) = 2x - 3. Find f(3).

L37-P012 | Let f(x) = 3x - 2. Find f(-2).

L37-P013 | Let f(x) = 4x - 1. Find f(0).

L37-P014 | Let f(x) = 5x + 0. Find f(2).

L37-P015 | Let f(x) = 6x + 1. Find f(-3).

L37-P016 | Let f(x) = 2x + 2. Find f(-1).

L37-P017 | Let f(x) = 3x + 3. Find f(1).

L37-P018 | Let f(x) = 4x + 4. Find f(3).

L37-P019 | Let f(x) = 5x - 4. Find f(-2).

L37-P020 | Let f(x) = 6x - 3. Find f(0).

2/5 | Practitioner

L37-P021 | f(x) = 1x^2 - 4x + 1. Evaluate f(-3).

L37-P022 | f(x) = 2x^2 - 3x + 2. Evaluate f(-1).

L37-P023 | f(x) = 3x^2 - 2x + 3. Evaluate f(1).

L37-P024 | f(x) = 1x^2 - 1x + 4. Evaluate f(3).

L37-P025 | f(x) = 2x^2 + 0x + 1. Evaluate f(-2).

L37-P026 | f(x) = 3x^2 + 1x + 2. Evaluate f(0).

L37-P027 | f(x) = 1x^2 + 2x + 3. Evaluate f(2).

L37-P028 | f(x) = 2x^2 + 3x + 4. Evaluate f(-3).

L37-P029 | f(x) = 3x^2 + 4x + 1. Evaluate f(-1).

L37-P030 | f(x) = 1x^2 - 4x + 2. Evaluate f(1).

L37-P031 | f(x) = 2x^2 - 3x + 3. Evaluate f(3).

L37-P032 | f(x) = 3x^2 - 2x + 4. Evaluate f(-2).

L37-P033 | f(x) = 1x^2 - 1x + 1. Evaluate f(0).

L37-P034 | f(x) = 2x^2 + 0x + 2. Evaluate f(2).

L37-P035 | f(x) = 3x^2 + 1x + 3. Evaluate f(-3).

L37-P036 | f(x) = 1x^2 + 2x + 4. Evaluate f(-1).

L37-P037 | f(x) = 2x^2 + 3x + 1. Evaluate f(1).

L37-P038 | f(x) = 3x^2 + 4x + 2. Evaluate f(3).

L37-P039 | f(x) = 1x^2 - 4x + 3. Evaluate f(-2).

L37-P040 | f(x) = 2x^2 - 3x + 4. Evaluate f(0).

3/5 | Challenger

L37-P041 | Find the domain of f(x) = (x^2 + 1)/((x - 2)(x + 1)).

L37-P042 | Find the domain of f(x) = (x^2 + 2)/((x - 3)(x + 2)).

L37-P043 | Find the domain of f(x) = (x^2 + 3)/((x - 4)(x + 3)).

L37-P044 | Find the domain of f(x) = (x^2 + 4)/((x - 5)(x + 4)).

L37-P045 | Find the domain of f(x) = (x^2 + 1)/((x - 6)(x + 1)).

L37-P046 | Find the domain of f(x) = (x^2 + 2)/((x - 2)(x + 2)).

L37-P047 | Find the domain of f(x) = (x^2 + 3)/((x - 3)(x + 3)).

L37-P048 | Find the domain of f(x) = (x^2 + 4)/((x - 4)(x + 4)).

L37-P049 | Find the domain of f(x) = (x^2 + 1)/((x - 5)(x + 1)).

L37-P050 | Find the domain of f(x) = (x^2 + 2)/((x - 6)(x + 2)).

L37-P051 | Find the domain of f(x) = (x^2 + 3)/((x - 2)(x + 3)).

L37-P052 | Find the domain of f(x) = (x^2 + 4)/((x - 3)(x + 4)).

L37-P053 | Find the domain of f(x) = (x^2 + 1)/((x - 4)(x + 1)).

L37-P054 | Find the domain of f(x) = (x^2 + 2)/((x - 5)(x + 2)).

L37-P055 | Find the domain of f(x) = (x^2 + 3)/((x - 6)(x + 3)).

L37-P056 | Find the domain of f(x) = (x^2 + 4)/((x - 2)(x + 4)).

L37-P057 | Find the domain of f(x) = (x^2 + 1)/((x - 3)(x + 1)).

L37-P058 | Find the domain of f(x) = (x^2 + 2)/((x - 4)(x + 2)).

L37-P059 | Find the domain of f(x) = (x^2 + 3)/((x - 5)(x + 3)).

L37-P060 | Find the domain of f(x) = (x^2 + 4)/((x - 6)(x + 4)).

4/5 | Expert

L37-P061 | f(x) = 2x - 4. Solve f(x) = -10 and state the corresponding graph point.

L37-P062 | f(x) = 3x - 3. Solve f(x) = -6 and state the corresponding graph point.

L37-P063 | f(x) = 4x - 2. Solve f(x) = 2 and state the corresponding graph point.

L37-P064 | f(x) = 5x - 1. Solve f(x) = 14 and state the corresponding graph point.

L37-P065 | f(x) = 6x + 0. Solve f(x) = -12 and state the corresponding graph point.

L37-P066 | f(x) = 2x + 1. Solve f(x) = 1 and state the corresponding graph point.

L37-P067 | f(x) = 3x + 2. Solve f(x) = 8 and state the corresponding graph point.

L37-P068 | f(x) = 4x + 3. Solve f(x) = -9 and state the corresponding graph point.

L37-P069 | f(x) = 5x + 4. Solve f(x) = -1 and state the corresponding graph point.

L37-P070 | f(x) = 6x - 4. Solve f(x) = 2 and state the corresponding graph point.

L37-P071 | f(x) = 2x - 3. Solve f(x) = 3 and state the corresponding graph point.

L37-P072 | f(x) = 3x - 2. Solve f(x) = -8 and state the corresponding graph point.

L37-P073 | f(x) = 4x - 1. Solve f(x) = -1 and state the corresponding graph point.

L37-P074 | f(x) = 5x + 0. Solve f(x) = 10 and state the corresponding graph point.

L37-P075 | f(x) = 6x + 1. Solve f(x) = -17 and state the corresponding graph point.

L37-P076 | f(x) = 2x + 2. Solve f(x) = 0 and state the corresponding graph point.

L37-P077 | f(x) = 3x + 3. Solve f(x) = 6 and state the corresponding graph point.

L37-P078 | f(x) = 4x + 4. Solve f(x) = 16 and state the corresponding graph point.

L37-P079 | f(x) = 5x - 4. Solve f(x) = -14 and state the corresponding graph point.

L37-P080 | f(x) = 6x - 3. Solve f(x) = -3 and state the corresponding graph point.

5/5 | Ninja

L37-P081 | Let g(x) = 2x + 1 and f(u) = 1/(u - 2). Find the domain of f∘g, then (f∘g)(1).

L37-P082 | Let g(x) = 3x + 2 and f(u) = 1/(u - 3). Find the domain of f∘g, then (f∘g)(2).

L37-P083 | Let g(x) = 4x + 3 and f(u) = 1/(u - 4). Find the domain of f∘g, then (f∘g)(3).

L37-P084 | Let g(x) = 5x + 1 and f(u) = 1/(u - 5). Find the domain of f∘g, then (f∘g)(4).

L37-P085 | Let g(x) = 2x + 2 and f(u) = 1/(u - 2). Find the domain of f∘g, then (f∘g)(1).

L37-P086 | Let g(x) = 3x + 3 and f(u) = 1/(u - 3). Find the domain of f∘g, then (f∘g)(2).

L37-P087 | Let g(x) = 4x + 1 and f(u) = 1/(u - 4). Find the domain of f∘g, then (f∘g)(3).

L37-P088 | Let g(x) = 5x + 2 and f(u) = 1/(u - 5). Find the domain of f∘g, then (f∘g)(4).

L37-P089 | Let g(x) = 2x + 3 and f(u) = 1/(u - 2). Find the domain of f∘g, then (f∘g)(1).

L37-P090 | Let g(x) = 3x + 1 and f(u) = 1/(u - 3). Find the domain of f∘g, then (f∘g)(2).

L37-P091 | Let g(x) = 4x + 2 and f(u) = 1/(u - 4). Find the domain of f∘g, then (f∘g)(3).

L37-P092 | Let g(x) = 5x + 3 and f(u) = 1/(u - 5). Find the domain of f∘g, then (f∘g)(4).

L37-P093 | Let g(x) = 2x + 1 and f(u) = 1/(u - 2). Find the domain of f∘g, then (f∘g)(1).

L37-P094 | Let g(x) = 3x + 2 and f(u) = 1/(u - 3). Find the domain of f∘g, then (f∘g)(2).

L37-P095 | Let g(x) = 4x + 3 and f(u) = 1/(u - 4). Find the domain of f∘g, then (f∘g)(3).

L37-P096 | Let g(x) = 5x + 1 and f(u) = 1/(u - 5). Find the domain of f∘g, then (f∘g)(4).

L37-P097 | Let g(x) = 2x + 2 and f(u) = 1/(u - 2). Find the domain of f∘g, then (f∘g)(1).

L37-P098 | Let g(x) = 3x + 3 and f(u) = 1/(u - 3). Find the domain of f∘g, then (f∘g)(2).

L37-P099 | Let g(x) = 4x + 1 and f(u) = 1/(u - 4). Find the domain of f∘g, then (f∘g)(3).

L37-P100 | Let g(x) = 5x + 2 and f(u) = 1/(u - 5). Find the domain of f∘g, then (f∘g)(4).

L38 | Average rates, derivatives and tangents

Source: PDF pages 5, 6, 7, 8, 9, 10, 11, 12

1/5 | Apprentice

L38-P001 | For f(t) = 1x - 3, find (f(-1) - f(-2))/1.

L38-P002 | For f(t) = 2x - 2, find (f(1) - f(-1))/2.

L38-P003 | For f(t) = 3x - 1, find (f(3) - f(0))/3.

L38-P004 | For f(t) = 4x + 0, find (f(2) - f(1))/1.

L38-P005 | For f(t) = 1x + 1, find (f(4) - f(2))/2.

L38-P006 | For f(t) = 2x + 2, find (f(1) - f(-2))/3.

L38-P007 | For f(t) = 3x + 3, find (f(0) - f(-1))/1.

L38-P008 | For f(t) = 4x - 3, find (f(2) - f(0))/2.

L38-P009 | For f(t) = 1x - 2, find (f(4) - f(1))/3.

L38-P010 | For f(t) = 2x - 1, find (f(3) - f(2))/1.

L38-P011 | For f(t) = 3x + 0, find (f(0) - f(-2))/2.

L38-P012 | For f(t) = 4x + 1, find (f(2) - f(-1))/3.

L38-P013 | For f(t) = 1x + 2, find (f(1) - f(0))/1.

L38-P014 | For f(t) = 2x + 3, find (f(3) - f(1))/2.

L38-P015 | For f(t) = 3x - 3, find (f(5) - f(2))/3.

L38-P016 | For f(t) = 4x - 2, find (f(-1) - f(-2))/1.

L38-P017 | For f(t) = 1x - 1, find (f(1) - f(-1))/2.

L38-P018 | For f(t) = 2x + 0, find (f(3) - f(0))/3.

L38-P019 | For f(t) = 3x + 1, find (f(2) - f(1))/1.

L38-P020 | For f(t) = 4x + 2, find (f(4) - f(2))/2.

2/5 | Practitioner

L38-P021 | For f(x) = 1x^2, compute the secant slope from x = -2 to x = -1.

L38-P022 | For f(x) = 2x^2, compute the secant slope from x = -1 to x = 1.

L38-P023 | For f(x) = 3x^2, compute the secant slope from x = 0 to x = 3.

L38-P024 | For f(x) = 4x^2, compute the secant slope from x = 1 to x = 2.

L38-P025 | For f(x) = 1x^2, compute the secant slope from x = 2 to x = 4.

L38-P026 | For f(x) = 2x^2, compute the secant slope from x = -2 to x = 1.

L38-P027 | For f(x) = 3x^2, compute the secant slope from x = -1 to x = 0.

L38-P028 | For f(x) = 4x^2, compute the secant slope from x = 0 to x = 2.

L38-P029 | For f(x) = 1x^2, compute the secant slope from x = 1 to x = 4.

L38-P030 | For f(x) = 2x^2, compute the secant slope from x = 2 to x = 3.

L38-P031 | For f(x) = 3x^2, compute the secant slope from x = -2 to x = 0.

L38-P032 | For f(x) = 4x^2, compute the secant slope from x = -1 to x = 2.

L38-P033 | For f(x) = 1x^2, compute the secant slope from x = 0 to x = 1.

L38-P034 | For f(x) = 2x^2, compute the secant slope from x = 1 to x = 3.

L38-P035 | For f(x) = 3x^2, compute the secant slope from x = 2 to x = 5.

L38-P036 | For f(x) = 4x^2, compute the secant slope from x = -2 to x = -1.

L38-P037 | For f(x) = 1x^2, compute the secant slope from x = -1 to x = 1.

L38-P038 | For f(x) = 2x^2, compute the secant slope from x = 0 to x = 3.

L38-P039 | For f(x) = 3x^2, compute the secant slope from x = 1 to x = 2.

L38-P040 | For f(x) = 4x^2, compute the secant slope from x = 2 to x = 4.

3/5 | Challenger

L38-P041 | Using the difference quotient limit, find f’(-2) for f(x) = 1x^2 - 3x.

L38-P042 | Using the difference quotient limit, find f’(-1) for f(x) = 2x^2 - 2x.

L38-P043 | Using the difference quotient limit, find f’(0) for f(x) = 3x^2 - 1x.

L38-P044 | Using the difference quotient limit, find f’(1) for f(x) = 4x^2 + 0x.

L38-P045 | Using the difference quotient limit, find f’(2) for f(x) = 1x^2 + 1x.

L38-P046 | Using the difference quotient limit, find f’(-2) for f(x) = 2x^2 + 2x.

L38-P047 | Using the difference quotient limit, find f’(-1) for f(x) = 3x^2 + 3x.

L38-P048 | Using the difference quotient limit, find f’(0) for f(x) = 4x^2 - 3x.

L38-P049 | Using the difference quotient limit, find f’(1) for f(x) = 1x^2 - 2x.

L38-P050 | Using the difference quotient limit, find f’(2) for f(x) = 2x^2 - 1x.

L38-P051 | Using the difference quotient limit, find f’(-2) for f(x) = 3x^2 + 0x.

L38-P052 | Using the difference quotient limit, find f’(-1) for f(x) = 4x^2 + 1x.

L38-P053 | Using the difference quotient limit, find f’(0) for f(x) = 1x^2 + 2x.

L38-P054 | Using the difference quotient limit, find f’(1) for f(x) = 2x^2 + 3x.

L38-P055 | Using the difference quotient limit, find f’(2) for f(x) = 3x^2 - 3x.

L38-P056 | Using the difference quotient limit, find f’(-2) for f(x) = 4x^2 - 2x.

L38-P057 | Using the difference quotient limit, find f’(-1) for f(x) = 1x^2 - 1x.

L38-P058 | Using the difference quotient limit, find f’(0) for f(x) = 2x^2 + 0x.

L38-P059 | Using the difference quotient limit, find f’(1) for f(x) = 3x^2 + 1x.

L38-P060 | Using the difference quotient limit, find f’(2) for f(x) = 4x^2 + 2x.

4/5 | Expert

L38-P061 | f(x) = 1x^2 - 3x. Find the tangent-line equation at x = -2.

L38-P062 | f(x) = 2x^2 - 2x. Find the tangent-line equation at x = -1.

L38-P063 | f(x) = 3x^2 - 1x. Find the tangent-line equation at x = 0.

L38-P064 | f(x) = 4x^2 + 0x. Find the tangent-line equation at x = 1.

L38-P065 | f(x) = 1x^2 + 1x. Find the tangent-line equation at x = 2.

L38-P066 | f(x) = 2x^2 + 2x. Find the tangent-line equation at x = -2.

L38-P067 | f(x) = 3x^2 + 3x. Find the tangent-line equation at x = -1.

L38-P068 | f(x) = 4x^2 - 3x. Find the tangent-line equation at x = 0.

L38-P069 | f(x) = 1x^2 - 2x. Find the tangent-line equation at x = 1.

L38-P070 | f(x) = 2x^2 - 1x. Find the tangent-line equation at x = 2.

L38-P071 | f(x) = 3x^2 + 0x. Find the tangent-line equation at x = -2.

L38-P072 | f(x) = 4x^2 + 1x. Find the tangent-line equation at x = -1.

L38-P073 | f(x) = 1x^2 + 2x. Find the tangent-line equation at x = 0.

L38-P074 | f(x) = 2x^2 + 3x. Find the tangent-line equation at x = 1.

L38-P075 | f(x) = 3x^2 - 3x. Find the tangent-line equation at x = 2.

L38-P076 | f(x) = 4x^2 - 2x. Find the tangent-line equation at x = -2.

L38-P077 | f(x) = 1x^2 - 1x. Find the tangent-line equation at x = -1.

L38-P078 | f(x) = 2x^2 + 0x. Find the tangent-line equation at x = 0.

L38-P079 | f(x) = 3x^2 + 1x. Find the tangent-line equation at x = 1.

L38-P080 | f(x) = 4x^2 + 2x. Find the tangent-line equation at x = 2.

5/5 | Ninja

L38-P081 | From the definition, differentiate f(x) = 1x^3 - 3x^2 - 1x at x = -2; find its tangent.

L38-P082 | From the definition, differentiate f(x) = 2x^3 - 2x^2 + 0x at x = -1; find its tangent.

L38-P083 | From the definition, differentiate f(x) = 3x^3 - 1x^2 + 1x at x = 0; find its tangent.

L38-P084 | From the definition, differentiate f(x) = 4x^3 + 0x^2 + 2x at x = 1; find its tangent.

L38-P085 | From the definition, differentiate f(x) = 1x^3 + 1x^2 - 1x at x = 2; find its tangent.

L38-P086 | From the definition, differentiate f(x) = 2x^3 + 2x^2 + 0x at x = -2; find its tangent.

L38-P087 | From the definition, differentiate f(x) = 3x^3 + 3x^2 + 1x at x = -1; find its tangent.

L38-P088 | From the definition, differentiate f(x) = 4x^3 - 3x^2 + 2x at x = 0; find its tangent.

L38-P089 | From the definition, differentiate f(x) = 1x^3 - 2x^2 - 1x at x = 1; find its tangent.

L38-P090 | From the definition, differentiate f(x) = 2x^3 - 1x^2 + 0x at x = 2; find its tangent.

L38-P091 | From the definition, differentiate f(x) = 3x^3 + 0x^2 + 1x at x = -2; find its tangent.

L38-P092 | From the definition, differentiate f(x) = 4x^3 + 1x^2 + 2x at x = -1; find its tangent.

L38-P093 | From the definition, differentiate f(x) = 1x^3 + 2x^2 - 1x at x = 0; find its tangent.

L38-P094 | From the definition, differentiate f(x) = 2x^3 + 3x^2 + 0x at x = 1; find its tangent.

L38-P095 | From the definition, differentiate f(x) = 3x^3 - 3x^2 + 1x at x = 2; find its tangent.

L38-P096 | From the definition, differentiate f(x) = 4x^3 - 2x^2 + 2x at x = -2; find its tangent.

L38-P097 | From the definition, differentiate f(x) = 1x^3 - 1x^2 - 1x at x = -1; find its tangent.

L38-P098 | From the definition, differentiate f(x) = 2x^3 + 0x^2 + 0x at x = 0; find its tangent.

L38-P099 | From the definition, differentiate f(x) = 3x^3 + 1x^2 + 1x at x = 1; find its tangent.

L38-P100 | From the definition, differentiate f(x) = 4x^3 + 2x^2 + 2x at x = 2; find its tangent.

L39 | Differentiation rules

Source: PDF pages 13

1/5 | Apprentice

L39-P001 | Differentiate f(x) = 1x^2.

L39-P002 | Differentiate f(x) = 2x^3.

L39-P003 | Differentiate f(x) = 3x^4.

L39-P004 | Differentiate f(x) = 4x^5.

L39-P005 | Differentiate f(x) = 5x^6.

L39-P006 | Differentiate f(x) = 1x^2.

L39-P007 | Differentiate f(x) = 2x^3.

L39-P008 | Differentiate f(x) = 3x^4.

L39-P009 | Differentiate f(x) = 4x^5.

L39-P010 | Differentiate f(x) = 5x^6.

L39-P011 | Differentiate f(x) = 1x^2.

L39-P012 | Differentiate f(x) = 2x^3.

L39-P013 | Differentiate f(x) = 3x^4.

L39-P014 | Differentiate f(x) = 4x^5.

L39-P015 | Differentiate f(x) = 5x^6.

L39-P016 | Differentiate f(x) = 1x^2.

L39-P017 | Differentiate f(x) = 2x^3.

L39-P018 | Differentiate f(x) = 3x^4.

L39-P019 | Differentiate f(x) = 4x^5.

L39-P020 | Differentiate f(x) = 5x^6.

2/5 | Practitioner

L39-P021 | f(x) = 1x^2 - 3x + 1. Find f’(-2).

L39-P022 | f(x) = 2x^2 - 2x + 2. Find f’(-1).

L39-P023 | f(x) = 3x^2 - 1x + 3. Find f’(0).

L39-P024 | f(x) = 4x^2 + 0x + 4. Find f’(1).

L39-P025 | f(x) = 5x^2 + 1x + 1. Find f’(2).

L39-P026 | f(x) = 1x^2 + 2x + 2. Find f’(-2).

L39-P027 | f(x) = 2x^2 + 3x + 3. Find f’(-1).

L39-P028 | f(x) = 3x^2 - 3x + 4. Find f’(0).

L39-P029 | f(x) = 4x^2 - 2x + 1. Find f’(1).

L39-P030 | f(x) = 5x^2 - 1x + 2. Find f’(2).

L39-P031 | f(x) = 1x^2 + 0x + 3. Find f’(-2).

L39-P032 | f(x) = 2x^2 + 1x + 4. Find f’(-1).

L39-P033 | f(x) = 3x^2 + 2x + 1. Find f’(0).

L39-P034 | f(x) = 4x^2 + 3x + 2. Find f’(1).

L39-P035 | f(x) = 5x^2 - 3x + 3. Find f’(2).

L39-P036 | f(x) = 1x^2 - 2x + 4. Find f’(-2).

L39-P037 | f(x) = 2x^2 - 1x + 1. Find f’(-1).

L39-P038 | f(x) = 3x^2 + 0x + 2. Find f’(0).

L39-P039 | f(x) = 4x^2 + 1x + 3. Find f’(1).

L39-P040 | f(x) = 5x^2 + 2x + 4. Find f’(2).

3/5 | Challenger

L39-P041 | Differentiate f(x) = (1x - 3)(1x - 3).

L39-P042 | Differentiate f(x) = (2x - 2)(2x + 0).

L39-P043 | Differentiate f(x) = (3x - 1)(3x + 3).

L39-P044 | Differentiate f(x) = (4x + 0)(4x - 1).

L39-P045 | Differentiate f(x) = (5x + 1)(1x + 2).

L39-P046 | Differentiate f(x) = (1x + 2)(2x - 2).

L39-P047 | Differentiate f(x) = (2x + 3)(3x + 1).

L39-P048 | Differentiate f(x) = (3x - 3)(4x - 3).

L39-P049 | Differentiate f(x) = (4x - 2)(1x + 0).

L39-P050 | Differentiate f(x) = (5x - 1)(2x + 3).

L39-P051 | Differentiate f(x) = (1x + 0)(3x - 1).

L39-P052 | Differentiate f(x) = (2x + 1)(4x + 2).

L39-P053 | Differentiate f(x) = (3x + 2)(1x - 2).

L39-P054 | Differentiate f(x) = (4x + 3)(2x + 1).

L39-P055 | Differentiate f(x) = (5x - 3)(3x - 3).

L39-P056 | Differentiate f(x) = (1x - 2)(4x + 0).

L39-P057 | Differentiate f(x) = (2x - 1)(1x + 3).

L39-P058 | Differentiate f(x) = (3x + 0)(2x - 1).

L39-P059 | Differentiate f(x) = (4x + 1)(3x + 2).

L39-P060 | Differentiate f(x) = (5x + 2)(4x - 2).

4/5 | Expert

L39-P061 | Differentiate f(x) = (x^2 + 2)/(x - 1), stating its domain.

L39-P062 | Differentiate f(x) = (x^2 + 3)/(x - 2), stating its domain.

L39-P063 | Differentiate f(x) = (x^2 + 4)/(x - 3), stating its domain.

L39-P064 | Differentiate f(x) = (x^2 + 5)/(x - 4), stating its domain.

L39-P065 | Differentiate f(x) = (x^2 + 6)/(x - 5), stating its domain.

L39-P066 | Differentiate f(x) = (x^2 + 7)/(x - 1), stating its domain.

L39-P067 | Differentiate f(x) = (x^2 + 2)/(x - 2), stating its domain.

L39-P068 | Differentiate f(x) = (x^2 + 3)/(x - 3), stating its domain.

L39-P069 | Differentiate f(x) = (x^2 + 4)/(x - 4), stating its domain.

L39-P070 | Differentiate f(x) = (x^2 + 5)/(x - 5), stating its domain.

L39-P071 | Differentiate f(x) = (x^2 + 6)/(x - 1), stating its domain.

L39-P072 | Differentiate f(x) = (x^2 + 7)/(x - 2), stating its domain.

L39-P073 | Differentiate f(x) = (x^2 + 2)/(x - 3), stating its domain.

L39-P074 | Differentiate f(x) = (x^2 + 3)/(x - 4), stating its domain.

L39-P075 | Differentiate f(x) = (x^2 + 4)/(x - 5), stating its domain.

L39-P076 | Differentiate f(x) = (x^2 + 5)/(x - 1), stating its domain.

L39-P077 | Differentiate f(x) = (x^2 + 6)/(x - 2), stating its domain.

L39-P078 | Differentiate f(x) = (x^2 + 7)/(x - 3), stating its domain.

L39-P079 | Differentiate f(x) = (x^2 + 2)/(x - 4), stating its domain.

L39-P080 | Differentiate f(x) = (x^2 + 3)/(x - 5), stating its domain.

5/5 | Ninja

L39-P081 | Use the chain rule to differentiate f(x) = (1x + 0)^2, then find f’(1).

L39-P082 | Use the chain rule to differentiate f(x) = (2x + 1)^3, then find f’(2).

L39-P083 | Use the chain rule to differentiate f(x) = (3x + 2)^4, then find f’(3).

L39-P084 | Use the chain rule to differentiate f(x) = (4x + 3)^5, then find f’(1).

L39-P085 | Use the chain rule to differentiate f(x) = (5x + 0)^2, then find f’(2).

L39-P086 | Use the chain rule to differentiate f(x) = (1x + 1)^3, then find f’(3).

L39-P087 | Use the chain rule to differentiate f(x) = (2x + 2)^4, then find f’(1).

L39-P088 | Use the chain rule to differentiate f(x) = (3x + 3)^5, then find f’(2).

L39-P089 | Use the chain rule to differentiate f(x) = (4x + 0)^2, then find f’(3).

L39-P090 | Use the chain rule to differentiate f(x) = (5x + 1)^3, then find f’(1).

L39-P091 | Use the chain rule to differentiate f(x) = (1x + 2)^4, then find f’(2).

L39-P092 | Use the chain rule to differentiate f(x) = (2x + 3)^5, then find f’(3).

L39-P093 | Use the chain rule to differentiate f(x) = (3x + 0)^2, then find f’(1).

L39-P094 | Use the chain rule to differentiate f(x) = (4x + 1)^3, then find f’(2).

L39-P095 | Use the chain rule to differentiate f(x) = (5x + 2)^4, then find f’(3).

L39-P096 | Use the chain rule to differentiate f(x) = (1x + 3)^5, then find f’(1).

L39-P097 | Use the chain rule to differentiate f(x) = (2x + 0)^2, then find f’(2).

L39-P098 | Use the chain rule to differentiate f(x) = (3x + 1)^3, then find f’(3).

L39-P099 | Use the chain rule to differentiate f(x) = (4x + 2)^4, then find f’(1).

L39-P100 | Use the chain rule to differentiate f(x) = (5x + 3)^5, then find f’(2).

L40 | The natural exponential function

Source: PDF pages 14, 15

1/5 | Apprentice

L40-P001 | Simplify e^2 × e^1 / e^0.

L40-P002 | Simplify e^3 × e^2 / e^1.

L40-P003 | Simplify e^4 × e^3 / e^2.

L40-P004 | Simplify e^5 × e^4 / e^3.

L40-P005 | Simplify e^6 × e^1 / e^4.

L40-P006 | Simplify e^2 × e^2 / e^0.

L40-P007 | Simplify e^3 × e^3 / e^1.

L40-P008 | Simplify e^4 × e^4 / e^2.

L40-P009 | Simplify e^5 × e^1 / e^3.

L40-P010 | Simplify e^6 × e^2 / e^4.

L40-P011 | Simplify e^2 × e^3 / e^0.

L40-P012 | Simplify e^3 × e^4 / e^1.

L40-P013 | Simplify e^4 × e^1 / e^2.

L40-P014 | Simplify e^5 × e^2 / e^3.

L40-P015 | Simplify e^6 × e^3 / e^4.

L40-P016 | Simplify e^2 × e^4 / e^0.

L40-P017 | Simplify e^3 × e^1 / e^1.

L40-P018 | Simplify e^4 × e^2 / e^2.

L40-P019 | Simplify e^5 × e^3 / e^3.

L40-P020 | Simplify e^6 × e^4 / e^4.

2/5 | Practitioner

L40-P021 | Compute f(0) and f’(0) for f(x) = 2e^x + 1.

L40-P022 | Compute f(0) and f’(0) for f(x) = 3e^x + 2.

L40-P023 | Compute f(0) and f’(0) for f(x) = 4e^x + 3.

L40-P024 | Compute f(0) and f’(0) for f(x) = 5e^x + 4.

L40-P025 | Compute f(0) and f’(0) for f(x) = 6e^x + 1.

L40-P026 | Compute f(0) and f’(0) for f(x) = 2e^x + 2.

L40-P027 | Compute f(0) and f’(0) for f(x) = 3e^x + 3.

L40-P028 | Compute f(0) and f’(0) for f(x) = 4e^x + 4.

L40-P029 | Compute f(0) and f’(0) for f(x) = 5e^x + 1.

L40-P030 | Compute f(0) and f’(0) for f(x) = 6e^x + 2.

L40-P031 | Compute f(0) and f’(0) for f(x) = 2e^x + 3.

L40-P032 | Compute f(0) and f’(0) for f(x) = 3e^x + 4.

L40-P033 | Compute f(0) and f’(0) for f(x) = 4e^x + 1.

L40-P034 | Compute f(0) and f’(0) for f(x) = 5e^x + 2.

L40-P035 | Compute f(0) and f’(0) for f(x) = 6e^x + 3.

L40-P036 | Compute f(0) and f’(0) for f(x) = 2e^x + 4.

L40-P037 | Compute f(0) and f’(0) for f(x) = 3e^x + 1.

L40-P038 | Compute f(0) and f’(0) for f(x) = 4e^x + 2.

L40-P039 | Compute f(0) and f’(0) for f(x) = 5e^x + 3.

L40-P040 | Compute f(0) and f’(0) for f(x) = 6e^x + 4.

3/5 | Challenger

L40-P041 | Differentiate f(x) = 2e^(1x + 1).

L40-P042 | Differentiate f(x) = 3e^(2x + 2).

L40-P043 | Differentiate f(x) = 4e^(3x + 3).

L40-P044 | Differentiate f(x) = 5e^(4x + 4).

L40-P045 | Differentiate f(x) = 6e^(1x + 1).

L40-P046 | Differentiate f(x) = 2e^(2x + 2).

L40-P047 | Differentiate f(x) = 3e^(3x + 3).

L40-P048 | Differentiate f(x) = 4e^(4x + 4).

L40-P049 | Differentiate f(x) = 5e^(1x + 1).

L40-P050 | Differentiate f(x) = 6e^(2x + 2).

L40-P051 | Differentiate f(x) = 2e^(3x + 3).

L40-P052 | Differentiate f(x) = 3e^(4x + 4).

L40-P053 | Differentiate f(x) = 4e^(1x + 1).

L40-P054 | Differentiate f(x) = 5e^(2x + 2).

L40-P055 | Differentiate f(x) = 6e^(3x + 3).

L40-P056 | Differentiate f(x) = 2e^(4x + 4).

L40-P057 | Differentiate f(x) = 3e^(1x + 1).

L40-P058 | Differentiate f(x) = 4e^(2x + 2).

L40-P059 | Differentiate f(x) = 5e^(3x + 3).

L40-P060 | Differentiate f(x) = 6e^(4x + 4).

4/5 | Expert

L40-P061 | Solve 2e^x = 2e^1 for real x.

L40-P062 | Solve 3e^x = 3e^2 for real x.

L40-P063 | Solve 4e^x = 4e^3 for real x.

L40-P064 | Solve 5e^x = 5e^4 for real x.

L40-P065 | Solve 6e^x = 6e^1 for real x.

L40-P066 | Solve 2e^x = 2e^2 for real x.

L40-P067 | Solve 3e^x = 3e^3 for real x.

L40-P068 | Solve 4e^x = 4e^4 for real x.

L40-P069 | Solve 5e^x = 5e^1 for real x.

L40-P070 | Solve 6e^x = 6e^2 for real x.

L40-P071 | Solve 2e^x = 2e^3 for real x.

L40-P072 | Solve 3e^x = 3e^4 for real x.

L40-P073 | Solve 4e^x = 4e^1 for real x.

L40-P074 | Solve 5e^x = 5e^2 for real x.

L40-P075 | Solve 6e^x = 6e^3 for real x.

L40-P076 | Solve 2e^x = 2e^4 for real x.

L40-P077 | Solve 3e^x = 3e^1 for real x.

L40-P078 | Solve 4e^x = 4e^2 for real x.

L40-P079 | Solve 5e^x = 5e^3 for real x.

L40-P080 | Solve 6e^x = 6e^4 for real x.

5/5 | Ninja

L40-P081 | A growth model is N(t)=2e^(1t); find N’(t), N’(0)/N(0), and interpret.

L40-P082 | A growth model is N(t)=3e^(2t); find N’(t), N’(1)/N(1), and interpret.

L40-P083 | A growth model is N(t)=4e^(3t); find N’(t), N’(2)/N(2), and interpret.

L40-P084 | A growth model is N(t)=5e^(4t); find N’(t), N’(3)/N(3), and interpret.

L40-P085 | A growth model is N(t)=6e^(1t); find N’(t), N’(0)/N(0), and interpret.

L40-P086 | A growth model is N(t)=2e^(2t); find N’(t), N’(1)/N(1), and interpret.

L40-P087 | A growth model is N(t)=3e^(3t); find N’(t), N’(2)/N(2), and interpret.

L40-P088 | A growth model is N(t)=4e^(4t); find N’(t), N’(3)/N(3), and interpret.

L40-P089 | A growth model is N(t)=5e^(1t); find N’(t), N’(0)/N(0), and interpret.

L40-P090 | A growth model is N(t)=6e^(2t); find N’(t), N’(1)/N(1), and interpret.

L40-P091 | A growth model is N(t)=2e^(3t); find N’(t), N’(2)/N(2), and interpret.

L40-P092 | A growth model is N(t)=3e^(4t); find N’(t), N’(3)/N(3), and interpret.

L40-P093 | A growth model is N(t)=4e^(1t); find N’(t), N’(0)/N(0), and interpret.

L40-P094 | A growth model is N(t)=5e^(2t); find N’(t), N’(1)/N(1), and interpret.

L40-P095 | A growth model is N(t)=6e^(3t); find N’(t), N’(2)/N(2), and interpret.

L40-P096 | A growth model is N(t)=2e^(4t); find N’(t), N’(3)/N(3), and interpret.

L40-P097 | A growth model is N(t)=3e^(1t); find N’(t), N’(0)/N(0), and interpret.

L40-P098 | A growth model is N(t)=4e^(2t); find N’(t), N’(1)/N(1), and interpret.

L40-P099 | A growth model is N(t)=5e^(3t); find N’(t), N’(2)/N(2), and interpret.

L40-P100 | A growth model is N(t)=6e^(4t); find N’(t), N’(3)/N(3), and interpret.

L41 | The natural logarithm

Source: PDF pages 16, 17

1/5 | Apprentice

L41-P001 | Compute ln(e^0) and e^(ln(2)).

L41-P002 | Compute ln(e^1) and e^(ln(3)).

L41-P003 | Compute ln(e^2) and e^(ln(4)).

L41-P004 | Compute ln(e^3) and e^(ln(5)).

L41-P005 | Compute ln(e^4) and e^(ln(6)).

L41-P006 | Compute ln(e^5) and e^(ln(2)).

L41-P007 | Compute ln(e^0) and e^(ln(3)).

L41-P008 | Compute ln(e^1) and e^(ln(4)).

L41-P009 | Compute ln(e^2) and e^(ln(5)).

L41-P010 | Compute ln(e^3) and e^(ln(6)).

L41-P011 | Compute ln(e^4) and e^(ln(2)).

L41-P012 | Compute ln(e^5) and e^(ln(3)).

L41-P013 | Compute ln(e^0) and e^(ln(4)).

L41-P014 | Compute ln(e^1) and e^(ln(5)).

L41-P015 | Compute ln(e^2) and e^(ln(6)).

L41-P016 | Compute ln(e^3) and e^(ln(2)).

L41-P017 | Compute ln(e^4) and e^(ln(3)).

L41-P018 | Compute ln(e^5) and e^(ln(4)).

L41-P019 | Compute ln(e^0) and e^(ln(5)).

L41-P020 | Compute ln(e^1) and e^(ln(6)).

2/5 | Practitioner

L41-P021 | Simplify ln(2) - ln(2); state the assumptions.

L41-P022 | Simplify ln(6) - ln(3); state the assumptions.

L41-P023 | Simplify ln(12) - ln(4); state the assumptions.

L41-P024 | Simplify ln(20) - ln(5); state the assumptions.

L41-P025 | Simplify ln(6) - ln(6); state the assumptions.

L41-P026 | Simplify ln(4) - ln(2); state the assumptions.

L41-P027 | Simplify ln(9) - ln(3); state the assumptions.

L41-P028 | Simplify ln(16) - ln(4); state the assumptions.

L41-P029 | Simplify ln(5) - ln(5); state the assumptions.

L41-P030 | Simplify ln(12) - ln(6); state the assumptions.

L41-P031 | Simplify ln(6) - ln(2); state the assumptions.

L41-P032 | Simplify ln(12) - ln(3); state the assumptions.

L41-P033 | Simplify ln(4) - ln(4); state the assumptions.

L41-P034 | Simplify ln(10) - ln(5); state the assumptions.

L41-P035 | Simplify ln(18) - ln(6); state the assumptions.

L41-P036 | Simplify ln(8) - ln(2); state the assumptions.

L41-P037 | Simplify ln(3) - ln(3); state the assumptions.

L41-P038 | Simplify ln(8) - ln(4); state the assumptions.

L41-P039 | Simplify ln(15) - ln(5); state the assumptions.

L41-P040 | Simplify ln(24) - ln(6); state the assumptions.

3/5 | Challenger

L41-P041 | For f(x)=ln(x), compute f’(1) and state the domain.

L41-P042 | For f(x)=ln(x), compute f’(2) and state the domain.

L41-P043 | For f(x)=ln(x), compute f’(3) and state the domain.

L41-P044 | For f(x)=ln(x), compute f’(4) and state the domain.

L41-P045 | For f(x)=ln(x), compute f’(1) and state the domain.

L41-P046 | For f(x)=ln(x), compute f’(2) and state the domain.

L41-P047 | For f(x)=ln(x), compute f’(3) and state the domain.

L41-P048 | For f(x)=ln(x), compute f’(4) and state the domain.

L41-P049 | For f(x)=ln(x), compute f’(1) and state the domain.

L41-P050 | For f(x)=ln(x), compute f’(2) and state the domain.

L41-P051 | For f(x)=ln(x), compute f’(3) and state the domain.

L41-P052 | For f(x)=ln(x), compute f’(4) and state the domain.

L41-P053 | For f(x)=ln(x), compute f’(1) and state the domain.

L41-P054 | For f(x)=ln(x), compute f’(2) and state the domain.

L41-P055 | For f(x)=ln(x), compute f’(3) and state the domain.

L41-P056 | For f(x)=ln(x), compute f’(4) and state the domain.

L41-P057 | For f(x)=ln(x), compute f’(1) and state the domain.

L41-P058 | For f(x)=ln(x), compute f’(2) and state the domain.

L41-P059 | For f(x)=ln(x), compute f’(3) and state the domain.

L41-P060 | For f(x)=ln(x), compute f’(4) and state the domain.

4/5 | Expert

L41-P061 | Differentiate f(x)=ln(2x + 1) and find f’(1); state the domain.

L41-P062 | Differentiate f(x)=ln(3x + 2) and find f’(2); state the domain.

L41-P063 | Differentiate f(x)=ln(4x + 3) and find f’(3); state the domain.

L41-P064 | Differentiate f(x)=ln(5x + 4) and find f’(4); state the domain.

L41-P065 | Differentiate $f(x)=\ln(6x + 1)$ and find $f'(1)$; state the domain.

L41-P066 | Differentiate $f(x)=\ln(2x + 2)$ and find $f'(2)$; state the domain.

L41-P067 | Differentiate $f(x)=\ln(3x + 3)$ and find $f'(3)$; state the domain.

L41-P068 | Differentiate $f(x)=\ln(4x + 4)$ and find $f'(4)$; state the domain.

L41-P069 | Differentiate $f(x)=\ln(5x + 1)$ and find $f'(1)$; state the domain.

L41-P070 | Differentiate $f(x)=\ln(6x + 2)$ and find $f'(2)$; state the domain.

L41-P071 | Differentiate $f(x)=\ln(2x + 3)$ and find $f'(3)$; state the domain.

L41-P072 | Differentiate $f(x)=\ln(3x + 4)$ and find $f'(4)$; state the domain.

L41-P073 | Differentiate $f(x)=\ln(4x + 1)$ and find $f'(1)$; state the domain.

L41-P074 | Differentiate $f(x)=\ln(5x + 2)$ and find $f'(2)$; state the domain.

L41-P075 | Differentiate $f(x)=\ln(6x + 3)$ and find $f'(3)$; state the domain.

L41-P076 | Differentiate $f(x)=\ln(2x + 4)$ and find $f'(4)$; state the domain.

L41-P077 | Differentiate $f(x)=\ln(3x + 1)$ and find $f'(1)$; state the domain.

L41-P078 | Differentiate f(x)=ln(4x + 2) and find f’(2); state the domain.

L41-P079 | Differentiate f(x)=ln(5x + 3) and find f’(3); state the domain.

L41-P080 | Differentiate f(x)=ln(6x + 4) and find f’(4); state the domain.

5/5 | Ninja

L41-P081 | Let f(x)=ln((2x + 1)^2). State the real domain and find f’(1).

L41-P082 | Let f(x)=ln((3x + 2)^3). State the real domain and find f’(2).

L41-P083 | Let f(x)=ln((4x + 3)^4). State the real domain and find f’(3).

L41-P084 | Let f(x)=ln((5x + 4)^5). State the real domain and find f’(4).

L41-P085 | Let f(x)=ln((6x + 1)^6). State the real domain and find f’(1).

L41-P086 | Let f(x)=ln((2x + 2)^2). State the real domain and find f’(2).

L41-P087 | Let f(x)=ln((3x + 3)^3). State the real domain and find f’(3).

L41-P088 | Let f(x)=ln((4x + 4)^4). State the real domain and find f’(4).

L41-P089 | Let f(x)=ln((5x + 1)^5). State the real domain and find f’(1).

L41-P090 | Let f(x)=ln((6x + 2)^6). State the real domain and find f’(2).

L41-P091 | Let f(x)=ln((2x + 3)^2). State the real domain and find f’(3).

L41-P092 | Let f(x)=ln((3x + 4)^3). State the real domain and find f’(4).

L41-P093 | Let f(x)=ln((4x + 1)^4). State the real domain and find f’(1).

L41-P094 | Let f(x)=ln((5x + 2)^5). State the real domain and find f’(2).

L41-P095 | Let f(x)=ln((6x + 3)^6). State the real domain and find f’(3).

L41-P096 | Let f(x)=ln((2x + 4)^2). State the real domain and find f’(4).

L41-P097 | Let f(x)=ln((3x + 1)^3). State the real domain and find f’(1).

L41-P098 | Let f(x)=ln((4x + 2)^4). State the real domain and find f’(2).

L41-P099 | Let f(x)=ln((5x + 3)^5). State the real domain and find f’(3).

L41-P100 | Let f(x)=ln((6x + 4)^6). State the real domain and find f’(4).

L42 | Bases, logarithms and complexity

Source: PDF pages 18, 19

1/5 | Apprentice

L42-P001 | Compute log_2(2^1) and 2^(log_2(2)).

L42-P002 | Compute log_3(3^2) and 3^(log_3(9)).

L42-P003 | Compute log_4(4^3) and 4^(log_4(64)).

L42-P004 | Compute log_5(5^4) and 5^(log_5(625)).

L42-P005 | Compute log_2(2^5) and 2^(log_2(32)).

L42-P006 | Compute log_3(3^6) and 3^(log_3(729)).

L42-P007 | Compute log_4(4^1) and 4^(log_4(4)).

L42-P008 | Compute log_5(5^2) and 5^(log_5(25)).

L42-P009 | Compute log_2(2^3) and 2^(log_2(8)).

L42-P010 | Compute log_3(3^4) and 3^(log_3(81)).

L42-P011 | Compute log_4(4^5) and 4^(log_4(1024)).

L42-P012 | Compute log_5(5^6) and 5^(log_5(15625)).

L42-P013 | Compute log_2(2^1) and 2^(log_2(2)).

L42-P014 | Compute log_3(3^2) and 3^(log_3(9)).

L42-P015 | Compute log_4(4^3) and 4^(log_4(64)).

L42-P016 | Compute log_5(5^4) and 5^(log_5(625)).

L42-P017 | Compute log_2(2^5) and 2^(log_2(32)).

L42-P018 | Compute log_3(3^6) and 3^(log_3(729)).

L42-P019 | Compute log_4(4^1) and 4^(log_4(4)).

L42-P020 | Compute log_5(5^2) and 5^(log_5(25)).

2/5 | Practitioner

L42-P021 | Differentiate f(x)=2^x; compute f’(0).

L42-P022 | Differentiate f(x)=3^x; compute f’(0).

L42-P023 | Differentiate f(x)=4^x; compute f’(0).

L42-P024 | Differentiate f(x)=5^x; compute f’(0).

L42-P025 | Differentiate f(x)=2^x; compute f’(0).

L42-P026 | Differentiate f(x)=3^x; compute f′(0).

L42-P027 | Differentiate f(x)=4^x; compute f′(0).

L42-P028 | Differentiate f(x)=5^x; compute f′(0).

L42-P029 | Differentiate f(x)=2^x; compute f′(0).

L42-P030 | Differentiate f(x)=3^x; compute f′(0).

L42-P031 | Differentiate f(x)=4^x; compute f′(0).

L42-P032 | Differentiate f(x)=5^x; compute f′(0).

L42-P033 | Differentiate f(x)=2^x; compute f′(0).

L42-P034 | Differentiate f(x)=3^x; compute f′(0).

L42-P035 | Differentiate f(x)=4^x; compute f′(0).

L42-P036 | Differentiate f(x)=5^x; compute f′(0).

L42-P037 | Differentiate f(x)=2^x; compute f′(0).

L42-P038 | Differentiate f(x)=3^x; compute f′(0).

L42-P039 | Differentiate f(x)=4^x; compute f’(0).

L42-P040 | Differentiate f(x)=5^x; compute f’(0).

3/5 | Challenger

L42-P041 | Find log_2(2) using change of base.

L42-P042 | Find log_3(9) using change of base.

L42-P043 | Find log_4(64) using change of base.

L42-P044 | Find log_5(625) using change of base.

L42-P045 | Find log_2(32) using change of base.

L42-P046 | Find log_3(729) using change of base.

L42-P047 | Find log_4(4) using change of base.

L42-P048 | Find log_5(25) using change of base.

L42-P049 | Find log_2(8) using change of base.

L42-P050 | Find log_3(81) using change of base.

L42-P051 | Find log_4(1024) using change of base.

L42-P052 | Find log_5(15625) using change of base.

L42-P053 | Find log_2(2) using change of base.

L42-P054 | Find log_3(9) using change of base.

L42-P055 | Find log_4(64) using change of base.

L42-P056 | Find log_5(625) using change of base.

L42-P057 | Find log_2(32) using change of base.

L42-P058 | Find log_3(729) using change of base.

L42-P059 | Find log_4(4) using change of base.

L42-P060 | Find log_5(25) using change of base.

4/5 | Expert

L42-P061 | Differentiate f(x)=log_2(2x + 2) and find f’(1); state the domain.

L42-P062 | Differentiate f(x)=log_3(3x + 3) and find f’(2); state the domain.

L42-P063 | Differentiate f(x)=log_4(4x + 4) and find f’(3); state the domain.

L42-P064 | Differentiate f(x)=log_5(5x + 2) and find f’(4); state the domain.

L42-P065 | Differentiate f(x)=log_2(2x + 3) and find f’(1); state the domain.

L42-P066 | Differentiate f(x)=log_3(3x + 4) and find f’(2); state the domain.

L42-P067 | Differentiate f(x)=log_4(4x + 2) and find f’(3); state the domain.

L42-P068 | Differentiate f(x)=log_5(5x + 3) and find f’(4); state the domain.

L42-P069 | Differentiate f(x)=log_2(2x + 4) and find f’(1); state the domain.

L42-P070 | Differentiate f(x)=log_3(3x + 2) and find f’(2); state the domain.

L42-P071 | Differentiate f(x)=log_4(4x + 3) and find f’(3); state the domain.

L42-P072 | Differentiate f(x)=log_5(5x + 4) and find f’(4); state the domain.

L42-P073 | Differentiate f(x)=log_2(2x + 2) and find f’(1); state the domain.

L42-P074 | Differentiate f(x)=log_3(3x + 3) and find f’(2); state the domain.

L42-P075 | Differentiate f(x)=log_4(4x + 4) and find f’(3); state the domain.

L42-P076 | Differentiate f(x)=log_5(5x + 2) and find f’(4); state the domain.

L42-P077 | Differentiate f(x)=log_2(2x + 3) and find f’(1); state the domain.

L42-P078 | Differentiate f(x)=log_3(3x + 4) and find f’(2); state the domain.

L42-P079 | Differentiate f(x)=log_4(4x + 2) and find f’(3); state the domain.

L42-P080 | Differentiate f(x)=log_5(5x + 3) and find f’(4); state the domain.

5/5 | Ninja

L42-P081 | At n=2^3=8, compare n $\log_2(n)$ with n^2. Which algorithm uses fewer operations at this n?

L42-P082 | At n=2^4=16, compare n $\log_2(n)$ with n^2. Which algorithm uses fewer operations at this n?

L42-P083 | At n=2^5=32, compare n $\log_2(n)$ with n^2. Which algorithm uses fewer operations at this n?

L42-P084 | At n=2^6=64, compare n $\log_2(n)$ with n^2. Which algorithm uses fewer operations at this n?

L42-P085 | At n=2^7=128, compare n $\log_2(n)$ with n^2. Which algorithm uses fewer operations at this n?

L42-P086 | At n=2^8=256, compare n $\log_2(n)$ with n^2. Which algorithm uses fewer operations at this n?

L42-P087 | At n=2^9=512, compare n $\log_2(n)$ with n^2. Which algorithm uses fewer operations at this n?

L42-P088 | At $n=2^3=8$, compare $n \log_2(n)$ with n^2. Which algorithm uses fewer operations at this n?

L42-P089 | At $n=2^4=16$, compare $n \log_2(n)$ with n^2. Which algorithm uses fewer operations at this n?

L42-P090 | At $n=2^5=32$, compare $n \log_2(n)$ with n^2. Which algorithm uses fewer operations at this n?

L42-P091 | At $n=2^6=64$, compare $n \log_2(n)$ with n^2. Which algorithm uses fewer operations at this n?

L42-P092 | At $n=2^7=128$, compare $n \log_2(n)$ with n^2. Which algorithm uses fewer operations at this n?

L42-P093 | At $n=2^8=256$, compare $n \log_2(n)$ with n^2. Which algorithm uses fewer operations at this n?

L42-P094 | At $n=2^9=512$, compare $n \log_2(n)$ with n^2. Which algorithm uses fewer operations at this n?

L42-P095 | At $n=2^3=8$, compare $n \log_2(n)$ with n^2. Which algorithm uses fewer operations at this n?

L42-P096 | At $n=2^4=16$, compare $n \log_2(n)$ with n^2. Which algorithm uses fewer operations at this n?

L42-P097 | At $n=2^5=32$, compare $n \log_2(n)$ with n^2. Which algorithm uses fewer operations at this n?

L42-P098 | At $n=2^6=64$, compare $n \log_2(n)$ with n^2. Which algorithm uses fewer operations at this n?

L42-P099 | At $n=2^7=128$, compare $n \log_2(n)$ with n^2. Which algorithm uses fewer operations at this n?

L42-P100 | At $n=2^8=256$, compare $n \log_2(n)$ with n^2. Which algorithm uses fewer operations at this n?

How to revise

Start with levels 1 and 2 without notes. Check the reasoning; retry mistakes after 48 hours. Levels 4 and 5 require selecting methods, not just longer calculations.